

Nazar Falach

Backend & Full-Stack Engineer

PROFILE

Backend and full-stack engineer building Node.js APIs, React and Next.js applications, PostgreSQL and MongoDB systems, Docker deployments, monitoring software, media pipelines, and Python AI tools.

CORE CAPABILITIES

Primary engineering

Node.js, Express, REST APIs, React, Next.js, TypeScript, database design, JWT, background workers

Data & operations

PostgreSQL, MongoDB, Prisma, Mongoose, Supabase, Docker Compose, Nginx, health checks

Applied AI & product

Python, PyTorch, Transformers, TRL, PEFT/LoRA, admin systems, monitoring, i18n, media processing

SELECTED ENGINEERING WORK

PulseGuard

API MONITORING

1s–24h checks · 5s worker tick · 10s timeout · 3-failure threshold

Per-user monitoring platform with endpoint lifecycle APIs, timestamped MongoDB logs, latency/uptime aggregation, and explicit UP/DOWN transitions.

- JWT and resource ownership checks
- Configurable intervals and expected status codes
- Next.js dashboard with Recharts trends

NEXT.JS · TYPESCRIPT · EXPRESS · MONGODB

Foundation Platform

CONTENT OPERATIONS

10 public routes · 6 RBAC levels · 5 Docker services · 3 domains

Bilingual public site and operational CMS with structured pages, publishing, partner management, user administration, audit history, and media handling.

- Server-authoritative RBAC and login throttling
- Compose health checks and persistent volumes
- Nginx routing for site, admin, and API

NEXT.JS · REACT · EXPRESS · PRISMA/SQL · DOCKER · NGINX

Prime Leather Repair

DEPLOYED CLIENT SYSTEM

45 MB media · 12h sessions · 4 tables · 2 storage buckets

Live service website and admin workflow for before/after media, categories, contacts, and reviews, backed by Supabase storage and data.

- HEIC conversion, image optimization, MOV/video support
- Scrypt passwords and HMAC-signed sessions
- Google Places and Resend integrations

REACT · VITE · EXPRESS · SUPABASE · RESEND

Local LLM Fine-Tuning

APPLIED AI R&D

11,590 examples · 1.1B model · LoRA r=8 · 3 epochs

Reproducible supervised fine-tuning pipeline that converts text to chat JSONL and trains parameter-efficient adapters on constrained hardware.

- CPU-safe float32 training profile
- Batch 1 with gradient accumulation 8
- Adapter and tokenizer artifact export

PYTHON · PYTORCH · TRANSFORMERS · TRL · PEFT